What does it mean to "ignore the overall phase-factor"? It means we can multiply any state-vector by a constant factor $e^{i\theta}$, where θ is a real number, without changing the state-vector's physical meaning. To see this, let's multiply Eq. 4.12 by $e^{i\theta}$ and call the result $|B\rangle$:

$$|B\rangle = e^{i\theta}|A\rangle = e^{i\theta}\sum_j \alpha_j|\lambda_j\rangle. \qquad (4.15)$$

Note that we changed the index in the summation from i to j to avoid confusion. It's easy to see that $|B\rangle$ has the same magnitude as $|A\rangle$, because $e^{i\theta}$ has a magnitude of one:

$$\langle B|B\rangle = \langle Ae^{-i\theta}|e^{i\theta}A\rangle = \langle A|A\rangle.$$

The same pattern of cancellation preserves other quantities as well. For example, $|A\rangle$'s probability amplitudes α_j become $e^{i\theta}\alpha_j$ for $|B\rangle$, so the probability *amplitudes* are different. However, it's the actual probability, not the amplitude, that has physical meaning. If a system is in state $|B\rangle$, and we perform a measurement, the result will be the eigenvalue of $|\lambda_j\rangle$ with probability

$$\alpha_j^* e^{-i\theta} e^{i\theta}\alpha_j = \alpha_j^*\alpha_j,$$

which is the same result we would get for state $|A\rangle$. Finally, let's use the same trick for the expectation value of a Hermitian operator **L**. Applying Eq. 4.14 to state $|B\rangle$, we can write

$$\langle \mathbf{L}\rangle = \langle B|\mathbf{L}|B\rangle.$$

Using Eq. 4.15 for $|B\rangle$, we get

$$\langle \mathbf{L}\rangle = \langle Ae^{-i\theta}|\mathbf{L}|e^{i\theta}A\rangle$$